

The Hong Kong Polytechnic University

Department of Computing

COMP4913 Capstone Project

Report (Interim)

Blind Drone Image Restoration Optimization with Real-ESRGAN

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1 Introduction

1.1 Background

Low-altitude economy is on the rise, and more drones will be used for inspection, mapping, monitoring, and situational awareness. The quality of drone images is becoming increasingly crucial, impacts both human decision-making and machine perception tasks such as OCR for signage/asset labels, defect detection, and object identification.

On the other hand, Real-ESRGAN has emerged as a widely adopted blind super-resolution (SR) method due to its robustness to mixed degradations: it uses ‘high-order degradation’ to create training data, to mimic complex real-world degradation factors, hence generating more realistic, higher quality, and sharper images.



Figure 1: Comparisons of bicubic-upsampled, ESRGAN, RealSR, and Real-ESRGAN results on real-life images. The Real-ESRGAN model trained with pure synthetic data is capable of enhancing details while removing annoying artifacts for common real-world images. [1]

However, domain-specific characteristics of aerial capture (platform motion, vibration, low-light conditions, and high dynamic range) expose limitations: persistent motion blur, difficulty in processing large images efficiently and artifact-free, and underperformance in dark regions where the signal-to-noise ratio (SNR) is low. Addressing these three challenges is essential to unlock reliable, high-quality enhancement for drone operations.

1.2 Problem Statement

1.2.1 Motion blur in low-altitude aerial capture

Cause: This uneven directional blur is a result of the motion of the drone and the motion of the moving objects, coupled with the exposure time, in other words, everything smudged, that is, the technical expression of all this. Geometry can be a problem with rolling-shutter effects, too, not only a blur issue, but a distortion issue.

Limit: Real-ESRGAN is not motion-aware, which means that the GAN will sharpen the image, but cannot consistently undo the blur. It frequently comes to hallucinate textures, and this is massive non-conformance in safety-critical applications in which you want the actual data.

Effects: The fact that the details are smudged (crack edges, wire strands, etc.) leads to a decrease in the interpretability of the imagery, and downstream tasks such as OCR or object detection are destroyed, in general the operational reliability is impaired.



Figure 2: The image on the left is the original input image, image on the right is the output image after being processed by Real-ESRGAN. The right image still has motion blurs.

1.2.2 Low-light poor performance and amplification of noise

Cause: Aerial shots (dusk, night, shadows) are low-light, with low SNR due to either shot/read noise or compression glitches; the contrast is so low that it is difficult to see textures.

Limit: Generic SR models have a tendency to enhance noises and form banding, or they over-smooth dark textures. The synthetic training data that real-ESRGAN works with does not even encompass those particular low-light sensor characteristics that are possessed by drones.

Impact: Dark areas remain fuzzy, OCR and defect detection is not reliable and analysts lose confidence due to the muddy or noisy outputs.

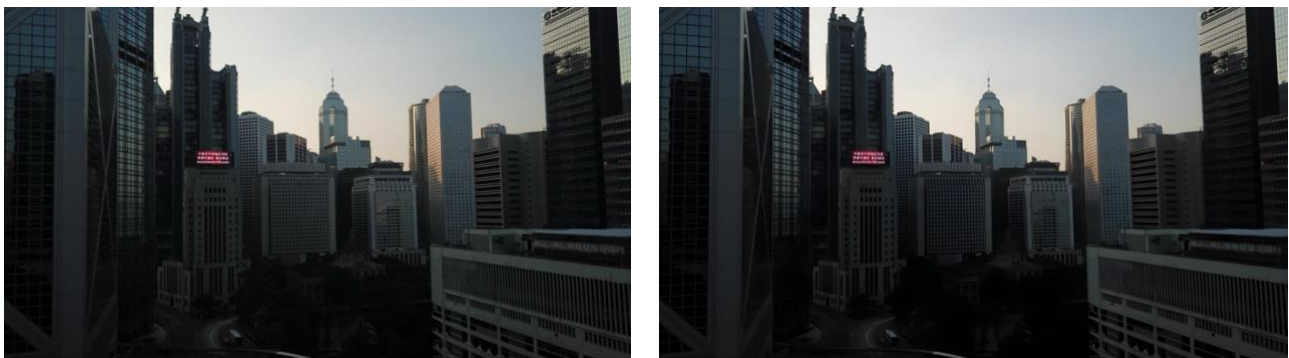


Figure 3: The image on the left is the original input image, image on the right is the output image after being processed by Real-ESRGAN. The right image shows indistinct in dark areas.

1.3 Objectives

This project aims to adapt and fine-tune Real-ESRGAN specifically for drone images, focusing on the following two measurable objectives.

1.3.1 Motion deblurring

This project aims to achieve a significant improvement on traditional methods(identity, bicubic, gaussian, sharpen, unsharp, and combined) and the pre-trained Real-ESRGAN model, which is $> +4$ dB PSNR improvement on traditional methods, $\geq +2$ dB PSNR on pre-trained Real-ESRGAN model, and $\geq 20\%$ improvement in no-reference sharpness metrics. For the primary evaluation metrics(with ground truth), we will use PSNR(Peak Signal-to-Noise Ratio), which measures pixel-level restoration accuracy, with higher the better; SSIM (Structural Similarity Index), which measures perceptual similarity, it has a range from 0 to 1, the higher the better; and LPIPS (Learned Perceptual Similarity), which is a deep learning-based perceptual metric, with range also 0 to 1, but the lower the better.

And for secondary evaluation metrics(no-reference sharpness metrics), we have variance of Laplacian, it convolves the image with a 3×3 Laplacian operator and calculates the variance, which is the squared standard deviation [2], to measure the second derivative of intensity of the picture, and emphasizes the areas of intense changes of the intensity. [3] And Tenengrad gradient function to evaluate the sharpness of the images. [4]

$$f(I) = \sum_{x=1}^M \sum_{y=1}^N \{ [I(x+1, y) - I(x, y)]^2 + [I(x, y+1) - I(x, y)]^2 \} \quad (3)$$

Figure 4: **Tenengrad function calculates the square of the difference of gray value between adjacent pixels. Where $I(x, y)$ is the gray value of iris image at pixel (x, y) ; M is the total row number of image; N is the total column number of image; $f(I)$ is the value of definition evaluation function. [4]**

1.3.2 Low-light enhancement with controlled artifacts

Low-light enhancement: $\geq +3$ dB PSNR improvement in dark regions (estimated via noise modeling or no-reference low-light metrics), $+0.05$ improvement on SSIM, and no visible halos/ringing artifacts. We also use PSNR, SSIM, and LPIPS, but also regional evaluation and noise suppression to have separate metrics calculation for shadow/dark areas and compare noise levels before and after enhancement for primary evaluation metrics.

And for secondary evaluation metrics, we originally planned to use NIQE and BRISQUE for no-reference evaluation, but since we have ground truth data and PSNR, SSIM provides a more

accurate measurement for the synthetic paired dataset, we replace them with luminance, entropy, and contrast, which are to measure the mean brightness improvement, information content in dark regions, and dynamic range restoration respectively.

1.4 Outcomes

1.4.1 Improved Clarity and Readability for Downstream Tasks

Sharper edges and restored details in previously blurred / motion-blurred areas, improved visibility of objects, especially vehicles and infrastructure, and enhanced visibility in dark regions with effective noise suppression

1.4.2 Reproducible Evaluation Suite and Comprehensive Baseline Analysis

Traditional methods including identity, bicubic, gaussian, sharpen, unsharp, combined, pre-trained Real-ESRGAN, and the fine-tuned Real-ESRGAN in the future. Optional OCR improvement if time permits.

1.5 Hardware, Software, and Libraries

GPU: at least Nvidia RTX 3060, CPU, RAM, storage: 25GB+

Python 3.9+, OpenCV 4.5+, NumPy 1.21+, PyTorch 1.12+ (with CUDA), scikit-image 0.19+, ultralytics (YOLOv8), lpips, pyiqa, Kornia, basicsr, pandas, pillow,

2 Literature Review

Real-ESRGAN is widely recognized as a strong perceptual super-resolution (SR) method and is often used “out of the box” to improve the apparent resolution and texture realism of natural images. UAV-related works follow the general trend of viewing Real-ESRGAN as a black-box solution to increment the recognition pipeline, instead of re-architecting it to match the physics of drone-specific restoration. For example, Yiğit et al. (2024) used Real-ESRGAN to enhance UAV imagery for object detection, reporting improvements over earlier GAN-based SR baselines (e.g., SRGAN).[11] While Bistrion et al. (2024) evaluated Real-ESRGAN on an aircraft-detection problem and found that SR can indeed be more efficient, but critical motion blur is a limiting case, and therefore, generic SR is not sufficient to overcome the prevalent blur failure modes in drone data.[12] There are also related Real-ESRGAN work non-standard drone modalities, such as thermal/infrared search-and-rescue imagery (e.g., Correa et al., 2025), but not directly solving low-light or motion blur restoration.[13] Existing UAV Real-ESRGAN literature supports its utility, but it generally uses the default pretrained weights and generic degradation assumptions, leaving performance under drone-specific motion dynamics and low-light noise insufficiently addressed.

There are also researches focus specifically on drone motion deblurring, such as Zhang et al. (2025) proposed an attention-guided deblurring Transformer for drone traffic surveillance, highlighting the importance of handling motion blur in practical UAV monitoring.[14] And another earlier work from SlimDeblurGAN (2020) addressed non-uniform blur for drone-related detection settings, demonstrating that blur characteristics in UAV imagery can be spatially varying and task-dependent.[15] However, these deblurring methods are not designed around Real-ESRGAN’s SR-and-restoration pipeline, so they may not deliver the same small-object

detail recovery associated with modern perceptual SR frameworks, and they are commonly optimized for deblurring alone rather than joint deblurring + low-light enhancement.

Although low-light drone images had also been studied such as a comparative study by Zhang et al. (2025) evaluating deep learning methods for super-resolving nighttime light images reported that Real-ESRGAN can perform well, but the study primarily assessed existing models rather than proposing drone-specific optimization.[5] And another research by Gan et al. (2024) explores enhancement in infrared/low-light settings[17]. They focus on denoising and visibility at night; they did not integrate SR, motion deblurring, and noise removal into a single blind restoration pipeline, which does not match the real-world drone scenario where low-light and motion blur occur at the same time.

Overall, the absence of integrated, drone-specific optimization of Real-ESRGAN for blind restoration forms the central research gap. This project addresses that gap by fine-tuning Real-ESRGAN for drone-specific degradations to improve motion deblurring and dark-region visibility while suppressing noise and artifacts.

3 Methodology

3.1 Datasets

For data collection, collect diverse representative drone images surroundings, altitudes, and compressions manually, then manually filter and extract high-quality images. We have tried to auto-curate images by using YOLOv8 and OpenCV to identify objects and use metrics to define images' quality, such as luminance and PSNR, however the result of image curating is not ideal, sometimes it filters out effective, satisfying, high quality images, sometimes it keeps the low quality, blurred images, so we filter images by hand, determined mainly by whether there are any motion blurs(sharpness) or brightness, resolution of the image is not our main concern since images could be taken from far distance so as it is not too bad we will still accept it. Then apply different degradations. Annotate subsets for motion severity, lighting, and presence of text.

This project will use multiple public UAV datasets to include diverse aerial scenes. Including VisDrone2019 dataset is collected by the AISKEYEYE team at the Lab of Machine Learning and Data Mining, Tianjin University, China, covering wide range of locations, environments, objects, and density in China;[6] FloodNet dataset provides over 10 landcover categories, such as background, water, building flooded, building non-flooded, road-flooded, etc.[7] However, in this project, the flood condition is not of our concern, so images will be filtered and only extract non-flooded images; Aerial-Traffic dataset provides Aerial Drone Traffic Image Dataset for Almaty City;[8] UAV-VisLoc provides a sequence of images on the UAV flight track;[9] And finally, the SwissOkutama provides Swiss Drone and Okutama Drone images Datasets.[10]

3.2 Data splitting

```

DATASET_FOLDERS = {
    "Aerial-Traffic": "Images_Aerial-Traffic",
    "FloodNet": "Images_FloodNet",
    "UAV-VisLoc": "Images_UAV-VisLoc",
    "VisDrone2019": "Images_VisDrone2019",
    "SwissOkutama": "Images_SwissOkutama"
}

SPLIT_RATIOS = {
    "train": 0.70, # 70%
    "val": 0.15, # 15%
    "test": 0.15 # 15%
}

RANDOM_SEED = 42

```


Figure 5: Dataset folders, data split ratios, and random seed

We use a data splitting strategy of 70% for training, 15% for validation, and 15% for testing, with the random seed 42 to ensure reproducibility.

3.3 Image Degradation

Since it is nearly impossible to find natural pairs of high-quality and low-quality images, we will add synthetic degradations to the high-quality images that we extracted, which in total 3 types of artifacts.

Global Motion Blur Synthesizer: which simulates the drone motion blur, we set the motion length to be randomly between 5 and 25 pixels, with random angles and kernel size of 15 for generating motion kernel, and apply the filter2D function from OpenCV, which uses FFT to each BGR channel to generate a blurred image.

Point Spread Function(PSF): $I_{\text{degraded}} = I_{\text{sharp}} * h + \eta$, h is the convolution, how it diffuse; η is noise

Object Motion Blur Synthesizer: which simulates object blurs in an image. We first use YOLOv8 nano version to identify the object vehicles, we use YOLOv8 because it is a more stable, simpler implementation, and faster version than the other newer versions, although the object detection result may be worse than the newer versions, but this is not our major concern, so we choose the simpler, faster version. Then to avoid hard edges, we applied Gaussian blur to feather mask edge. We have tried different confidence threshold for the YOLO which affects the accuracy and number of objects detected, and eventually chose to use the threshold of 0.2 to get more objects detected with fewer wrong detections, motion length between 5 and 15 pixels, also random angles, and 5 pixels for feather amount.

```
if YOLO_AVAILABLE:
    try:
        print("载入 YOLOv8-seg 模型...")
        self.model = YOLO('yolov8n-seg.pt')
        self.model.to(device)
        print(f"ObjectMotionBlurSynthesizer 使用设备: {device}")
    except Exception as e:
        print(f"载入 YOLOv8 模型失败: {e}")
        self.model = None

self.vehicle_classes = {
    2: 'car',
    3: 'motorcycle',
    5: 'bus',
    7: 'truck',
    1: 'bicycle',
}
```

Figure 6: Implementation of YOLOv8 and object classes

Low-light Degradation Synthesizer: The degradation includes multiple physical processes.

Image pipeline: $I_{\text{low-light}} = \text{fJPG}(\text{fnoise}(\text{ftone}(\text{fwb}(\text{fexp}(I_{\text{original}}))))$, from inside layer to outside layer functions are exposure reduction, white balance shift, apply tone curve/ tone mapping, add noise, and jpeg compression.

3.4 Baseline Check

We used 6 traditional methods to do a baseline check first, including identity which returns an unchanged image, bicubic which does simple bicubic interpolation (typically used for upscaling), unsharp masking which amplifies high-frequency details like edges, sharpen which convolves with a sharpening kernel, Gaussian blur, which applies Gaussian smoothing kernel, and combined which combines denoising and sharpening. We used several metrics to evaluate the

results of each method, as shown in the figures below. For LPIPS, we only evaluated 700 of them, as it is too time-consuming. And here are the expected results of each method:

Identity should be the baseline, a moderate score on everything, sharpness score should be low if the degraded input is blurry.

Bicubic should be similar to identity for PSNR and SSIM scores, and has a slight smoothing effect.

Sharpen is expected to have aggressive sharpening, the highest sharpness score, the worst PSNR and SSIM score since it creates artifacts and overshoots, and having low LPIPS score.

Unsharp should increase and have a higher Laplacian and tenengrad score, PSNR score might be decreased since it enhances noise, and might create artifacts.

Gaussian is expected to have low scores on sharpness metrics and SSIM since it loses details and has a moderate PSNR score as the result of smoothing will not create artifacts.

Combined have the potential of being the best among all, but could also have an average score on everything, as it is a combination of techniques.

method	var_laplacian	tenengrad	blur_extent	luminance	contrast	entropy	gradient_mean	psnr	ssim	lpips	reference_score	sharpness_score	blur_score	overall_score
sharpen	4797.6728	100.4205	0.2407	109.1007	45.5642	7.195	100.4205	20.5291	0.4293	0.497	0.2278775	1	0.67477368	0.47168124
unsharp	395.3129	41.433	0.449	108.9736	39.2233	6.9376	41.433	24.6513	0.6277	0.3841	0.4301325	0.24749592	0.39332523	0.38624373
bicubic	84.8495	28.7253	0.5927	108.9748	38.1789	6.8764	28.7253	25.1326	0.6458	0.4279	0.451215	0.151867856	0.19916228	0.34093503
identity	83.6185	28.563	0.5943	108.9742	38.1621	6.8756	28.563	25.1329	0.6457	0.4287	0.4511725	0.150931463	0.19700041	0.34028987
combined	138.5698	21.2878	0.5879	108.9624	38.0045	6.8454	21.2878	24.9657	0.6234	0.514	0.4358425	0.120434654	0.20564789	0.32672201
gaussian	12.0008	22.5133	0.7401	108.9758	37.7068	6.8499	22.5133	25.0923	0.637	0.5131	0.4458075	0.11334583	0	0.29015367

Figure 7: Metric scores of traditional methods on drone motion-blurred images, sorted by overall score

method	var_laplacian	tenengrad	blur_extent	luminance	contrast	entropy	gradient_mean	psnr	ssim	lpips	reference_score	sharpness_score	blur_score	overall_score
identity	995.156	62.8729	0.3799	110.7396	47.7515	7.2659	62.8729	37.4565	0.9805	0.0257	0.9266625	0.148077578	0.41625691	0.6688644
bicubic	996.6435	63.7163	0.3799	110.7388	47.8385	7.2686	63.7163	37.3049	0.9802	0.0261	0.9227225	0.149878435	0.41625691	0.66868057
unsharp	4685.9452	105.9645	0.2556	110.7193	52.7063	7.3951	105.9645	26.1687	0.8807	0.1627	0.5945675	0.296620251	0.60725261	0.53751507
combined	2607.5115	60.8292	0.306	110.6342	49.7175	7.268	60.8292	28.4815	0.883	0.2125	0.6535375	0.168985918	0.52980947	0.53188158
sharpen	31971.1981	237.2307	0.1373	112.1931	69.8572	7.2194	237.2307	15.3984	0.4603	0.6097	0.23015	1	0.78902889	0.49589578
gaussian	66.5935	40.6469	0.6508	110.7404	45.6523	7.1981	40.6469	28.5401	0.8703	0.276	0.6486525	0.086711022	0	0.4065337

Figure 8: Metric scores of traditional methods on object motion-blurred images, sorted by overall score

method	var_laplacian	tenengrad	blur_extent	luminance	contrast	entropy	gradient_mean	psnr	ssim	lpips	reference_score	sharpness_score	blur_score	overall_score
sharpen	9180.0501	138.8358	0.1078	48.3109	31.4448	6.5282	138.8358	11.3407	0.2947	0.9052	0.14735	1	0.81544256	0.45149851
unsharp	761.8415	48.8366	0.2121	46.0502	16.3315	5.8478	48.8366	11.7107	0.5022	0.5225	0.2511	0.217373397	0.63687725	0.32151013
bicubic	148.0072	27.0301	0.3269	46.0282	13.8308	5.5159	27.0301	11.6865	0.5204	0.4093	0.2602	0.10540692	0.44033556	0.2652685
identity	147.6565	26.8104	0.3272	46.0277	13.8082	5.513	26.8104	11.6851	0.5197	0.4101	0.25985	0.104596596	0.43982195	0.26479371
combined	60.0153	11.3105	0.4546	46.0113	12.8601	5.2714	11.3105	11.62	0.4495	0.551	0.22475	0.044002159	0.22170861	0.18799215
gaussian	10.0974	15.0082	0.5841	46.0274	12.8884	5.3278	15.0082	11.6327	0.4865	0.5284	0.24325	0.054600144	0	0.15687003

Figure 9: Metric scores of traditional methods on low-light images, sorted by overall score

The above 3 figures are the results of metric scores of the methods on different degraded images. For the special metrics, the formula of the reference score is $((\text{normalized psnr} + \text{normalized ssim}) / 2)$; the formula of the sharpness score is $((\text{Laplacian}/\text{max Laplacian}) + (\text{tenengrad}/\text{max tenengrad})) / 2$; and the overall score is $(\text{reference_score} \times 0.6 + \text{sharpness_score} \times 0.2 + \text{blur_score} \times 0.2)$.

For the actual results on the methods:

Identity has a very high PSNR and SSIM score on object motion blurred images; we believe that it is because the application of object motion blur only affects small parts of an image, while

having lower metric scores on drone motion blur and low-light images. It does have a moderate score among all methods.

Bicubic has almost identical results to identity as expected.

Sharpen has the highest score of Laplacian and tenengrad in almost all kinds of images, which leads to a maximum sharpness score, and worst results on PSNR, SSIM, and LPIPS, as expected.

Unsharp has a significantly higher score on Laplacian, tenengrad and sharpness score than other methods except for sharpen. The PSNR and SSIM score dropped sometimes even lower than identity. We believe that it is because the sharpening deviates from the ground truth image, and alters the structure.

Gaussian, as expected, has the lowest scores among all methods in all metrics, as it does the opposite of restoration.

Combined has a middle-ground performance, not the best and not the worst, not as expected to be the best, we think because it needs optimization to try out the best combinations.

Another part of the baseline check is to check the metric score of the pre-trained Real-ESRGAN model. For LPIPS, we also only evaluate 700 of them with a random seed.

method	var_laplacian	tenengrad	blur_extent	luminance	contrast	entropy	gradient_mean	psnr	ssim	lpips	reference_score	sharpness_score	blur_score	overall_score
realesrgan	878.1174	35.1655	0.3973	108.9776	39.5972	6.8994	35.1655	24.1816	0.6153	0.4275	0.41219	1	0	0.447314

Figure 10: **Metric scores of Real-ESRGAN on drone motion-blurred images**

method	var_laplacian	tenengrad	blur_extent	luminance	contrast	entropy	gradient_mean	psnr	ssim	lpips	reference_score	sharpness_score	blur_score	overall_score
realesrgan	3705.7949	64.3335	0.2766	110.487	51.0642	7.2835	64.3335	26.605	0.8582	0.1681	0.594225	1	0	0.556535

Figure 11: **Metric scores of Real-ESRGAN on object motion-blurred images**

method	var_laplacian	tenengrad	blur_extent	luminance	contrast	entropy	gradient_mean	psnr	ssim	lpips	reference_score	sharpness_score	blur_score	overall_score
realesrgan	310.7092	16.6899	0.2932	44.6008	14.2171	5.3857	16.6899	11.4331	0.4654	0.4647	0.2327	1	0	0.33962

Figure 12: **Metric scores of Real-ESRGAN on low-light images**

The above figures show the results of Real-ESRGAN, using the same metric used to evaluate the traditional methods. For the overall scores, we expected the result of object motion blurred images would be better than drone motion blurred images, and would be better than low-light images, and the actual results match our expectations. The result on object motion blurred images looks good because the blurs often only affect limited areas, while the surrounding regions are intact, and Real-ESRGAN is strong at hallucinating plausible high-frequency detail from local context and textures, so it can restore edges and textures that help SSIM and PSNR, and give low LPIPS.

The overall score on drone motion blur is expected and is actually lower than the score on object motion-blurred images, because drone motion blurs have longer and often directional smear across the whole image, it removes high-frequency content irreversibly. Single-image SR models have less signal to reconstruct correct fine detail, so they either underperform or hallucinate inconsistent textures. Hence, the PSNR and SSIM scores will drop, and LPIPS will be high because the perceptual content is different from the GT image.

It is expected and is performed the worst on low-light images, because low-light reduces SNR, producing heavy noises, quantization, color shifts, and clipped shadows. And Real-ESRGAN is not primarily used as a denoiser, so it may amplify noises, invent texture or even change the

intensity of the image, which all these effect lowers the PSNR score and worsen LPIPS, even if some edges might look sharper.

However, while comparing the Real-ESRGAN results with the traditional methods results, we can see that on object motion blurred images, identity and bicubic outperform Real-ESRGAN; on drone motion blurred images, sharpen outperforms Real-ESRGAN; on low-light images, sharpen also outperforms Real-ESRGAN. And here are the reasons that we believe cause this situation: for object motion-blurred images, object motion blur is often localized, and both identity and bicubic keep the original pixels, so they preserve the GT structures, while Real-ESRGAN might produce hallucinations like synthesized textures, shifted edges on blurred objects, which reduces PSNR and SSIM scores, so the overall score falls below these two methods. For drone motion-blurred images, above we have explained why sharpen have such a high overall score. Real-ESRGAN may attempt to recreate fine detail across a large directional smear, which produces inconsistent textures and lowers the related scores. For low-light images, they suffer from low SNR or color shifts, and sometimes sharpen raises perceptual sharpness lead to higher overall scores, while Real-ESRGAN may amplify noises, shift luminance and color even more, or produce unreal texture, leading lower overall score than sharpen. To conclude, the pre-trained Real-ESRGAN was not trained for these kinds of degradations, so the result might vary according to the type and how much an image degrades, and might not be able to beat the scores of traditional methods on some kinds of degraded images.

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